



Duckweed Cultivation Manual

A practical guide for new starters

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Prepared by

Dr Juanita Lauer

School of Agriculture, Food and Wine

Adelaide University

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1. Purpose and Scope

Duckweed (Lemnaceae) Cultivation Guide for Laboratory Use

This document provides a standardised, practical protocol for cultivating duckweed in a research laboratory setting. It is designed for new lab personnel, students, collaborators, and researchers needing reliable cultures for experiments such as growth studies, nutrient uptake, phytoremediation, biomass production, or wastewater applications.

Species Covered

This protocol spans five duckweed species:

- *Landoltia punctata*
- *Lemna aequinoctialis*
- *Lemna minuta*
- *Spirodela polyrhiza*
- *Wolffia australiana*

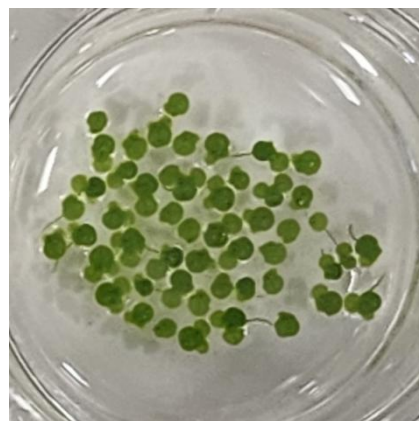
Duckweed species in order of size, left to right



Wolffia australiana



Wolffia australiana



Lemna minuta



Lemna aequinoctialis



Landoltia punctata



Spirodela polyrhiza

Starting Material

Cultures are typically received as sanitised or bleach-treated stock. This guide assumes you are beginning from this relatively clean state. If you need axenic (sterile) culture for a specific experiment, contact the Australian Collection of Duckweed Clones or Jenny Mortimer.

Cultivation Modes

Different experiments require different levels of cleanliness. This protocol covers two modes:

Semi-Sterile (recommended baseline for laboratory work): Cultures start clean (from sanitised stock and autoclaved media) and are maintained with good laboratory hygiene. Low levels of environmental microbes may gradually appear, but the duckweed remains dominant and healthy. This is the standard mode for most lab-based work.

Non-Sterile: Environmental or wastewater microbes are expected and tolerated. Primarily used for applied studies such as growth on anaerobic digestate or wastewater treatment trials.

Axenic (fully sterile) culture is used only for specialised experiments (e.g. microbiome studies, genetic work) requiring verified zero-microbial presence. This is managed through the Australian Collection of Duckweed Clones rather than covered in detail here — contact the Collection or Jenny Mortimer if your work requires it.

Recommended Workflow

- Maintain stock and experimental cultures in semi-sterile conditions in the laboratory to ensure consistency and reproducibility.
- For anaerobic digestate or wastewater applications, use healthy semi-sterile cultures as inoculum. Acclimate and scale them up in the lab first, then transfer into the target non-sterile medium. This staged approach improves establishment success.

Definition of Success

Success means sustained, vigorous growth with consistent morphology, predictable growth rates, and biomass suitable for downstream experiments. In semi-sterile mode this includes maintaining functionally clean cultures. In non-sterile applications, success is judged by the duckweed's ability to establish and grow robustly amid complex microbial communities.

2. What Success Looks Like

Healthy duckweed cultures show consistent, vigorous growth and uniform appearance. Regular visual comparison against a known healthy stock culture is the most reliable monitoring method.

Expected Growth Rate

Under standard semi-sterile lab conditions (20–28°C, 50–150 $\mu\text{mol m}^{-2} \text{s}^{-1}$ light, fresh media):

- Typical doubling time: 2–4 days (can be as fast as 1.5–3 days under optimal conditions).
- Slower growth is normal for some species.

Growth should be steady. Sudden stagnation or rapid crash usually indicates stress or contamination.

Morphology

Frond size, shape, and root characteristics are reliable identifiers for each species under healthy growth conditions.

- *Wolffia australiana*: Smallest species (0.3–1 mm) boat-shaped to ellipsoidal with a smooth, weakly convex upper surface. Rootless.

- *Lemna minuta*: 2–5 mm, oval to elliptical and flat; single root per frond. One of the fastest-growing species in the group.
- *Lemna aequinoctialis*: 1–6 mm, ovate to lance-shaped, often slightly asymmetrical; single root per frond. Daughter fronds frequently remain attached in small clusters.
- *Landoltia punctata*: 1–5 mm, narrowly egg-shaped to slightly kidney-shaped, intensely green with a waxy cuticle that gives a slight sparkle in light; 2–4 roots per frond (occasionally more).
- *Spirodela polyrhiza*: largest species, 5–10 mm, round to oval flat disc; green above; the clearest identifying feature is root count — 7–21 fine roots per frond, far more than any other species here.

Healthy fronds lie flat on the water surface without curling, folding, or distortion. Root presence and length (where applicable) vary with nutrient availability.

Surface Behaviour

- Even distribution: At lower densities, healthy cultures spread evenly across the surface, forming a single layer.
- Clumping: At higher densities, some clumping is normal but shouldn't form thick, multi-layered mats that block light and gas exchange. *Spirodela* and *Landoltia* tend to cluster more naturally than *Lemna*.
- Edge accumulation: Rather than clumping, *Wolffia* fronds stay mostly independent and are pushed toward the container edge by surface tension at the meniscus. This is not a sign of stress.

Visual Health Indicators

- Opacity: Healthy fronds appear solid and opaque. Stressed fronds look thinner or slightly translucent.
- Edge definition: Healthy = sharp, crisp edges. Stressed = softer, irregular, or frayed edges.
- Texture/Appearance: Healthy fronds look firm and flat. Stressed fronds can appear limp, wrinkled, or uneven.
- Colour intensity (secondary cue): Healthy = deep, saturated green. Stressed = pale, washed-out, or yellowish.

Species-Specific Stress Examples

Each plate shows a ½ Schenk & Hildebrandt (SH) control (top row) with healthy fronds alongside 20% dairy digestate (bottom row) with stressed fronds after three weeks of growth.

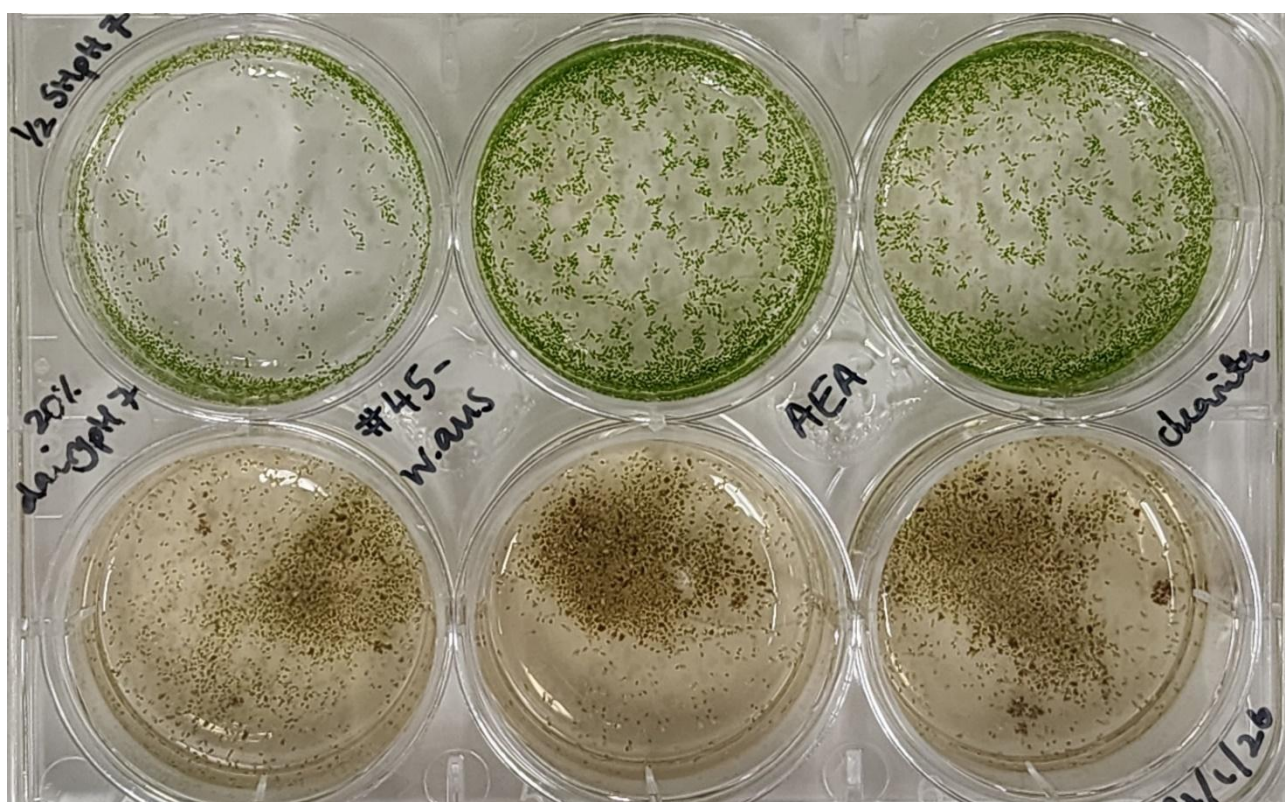
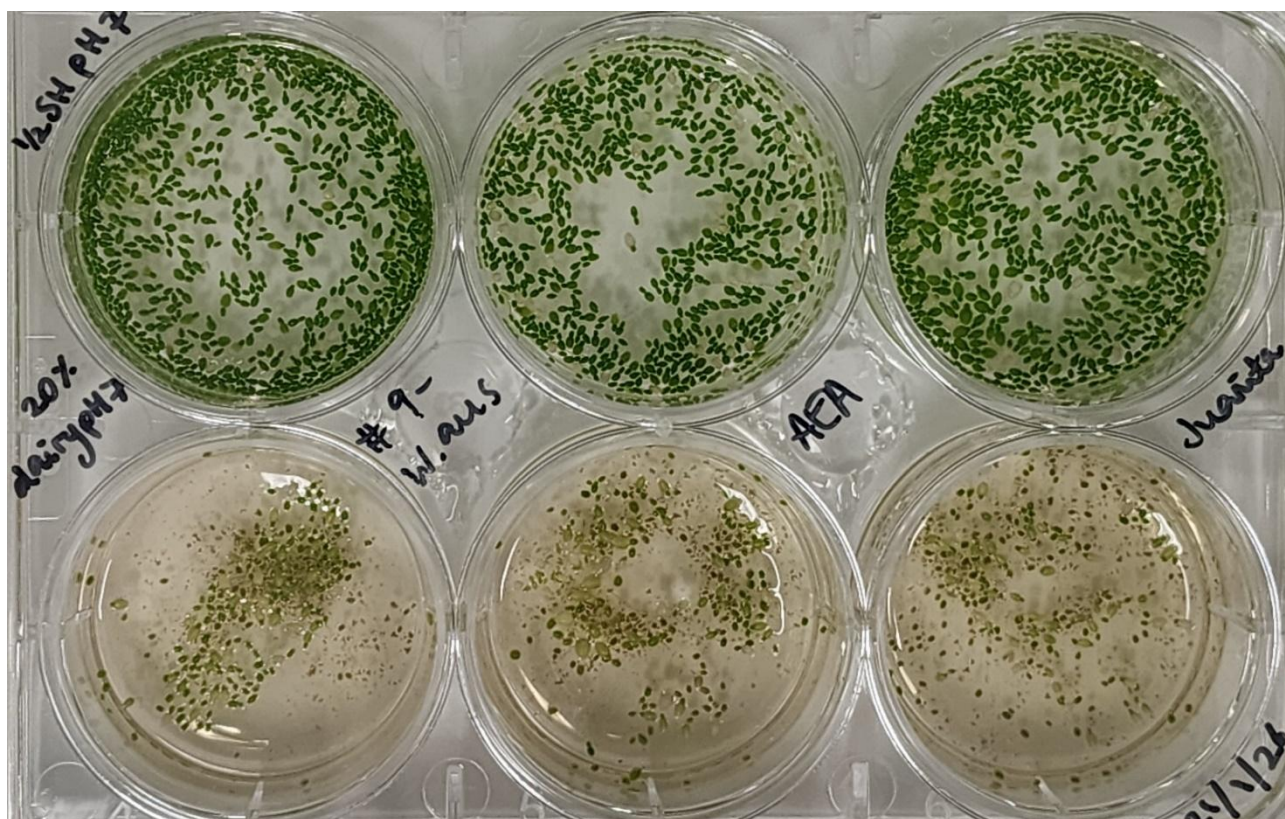
Stress indicators across the species included:

- Colour intensity: washed out pale green (*Landoltia punctata* and *Lemna minuta*) to pale yellow (*Lemna aequinoctialis* and *Spirodela polyrhiza*) to white (*Wolffia australiana*).
- Reduced surface coverage: less dense for all species, *Wolffia* also drop into suspension.
- Dormancy response: *Spirodela polyrhiza* produces turions as a survival mechanism in highly stressed conditions.

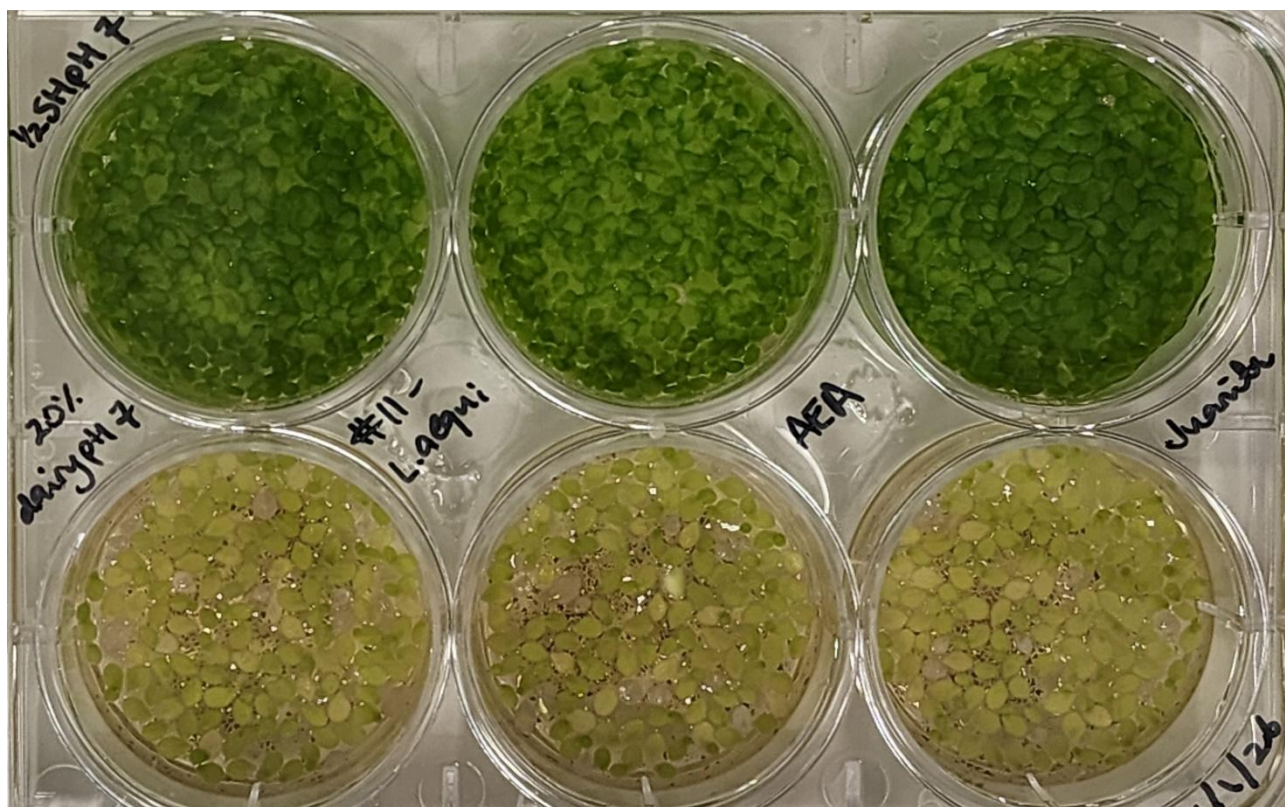
In extreme stress conditions, stunted growth patterns include shrunk fronds and abnormal growth (not shown in the examples below).

Overall, the strong performers were *Lemna minuta*, *Lemna aequinoctialis* and *Landoltia punctata*. The 20% dairy digestate growth media was largely lethal to *Wolffia australiana*, and *Spirodela polyrhiza* was stressed enough to trigger the dormancy response, producing turions.

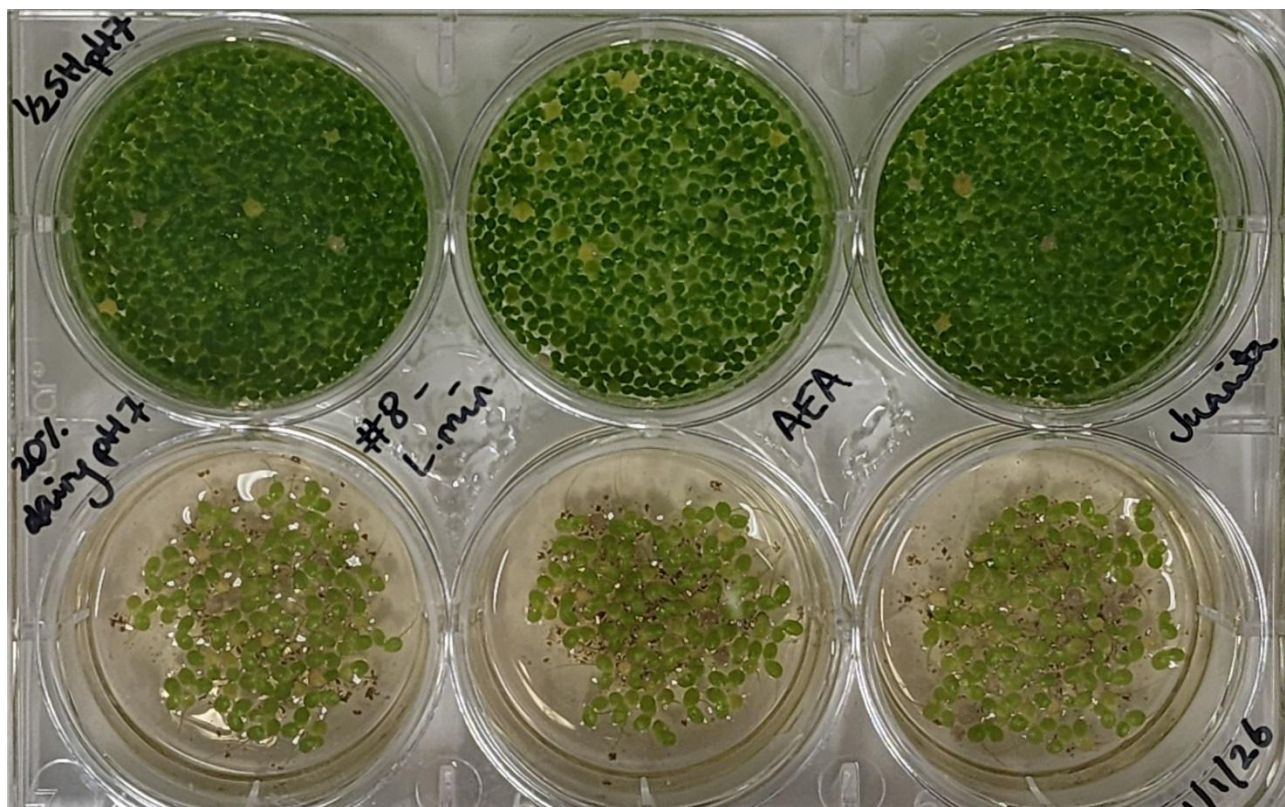
Wolffia australiana (two accessions shown — ~0.3 mm and ~1.0 mm)



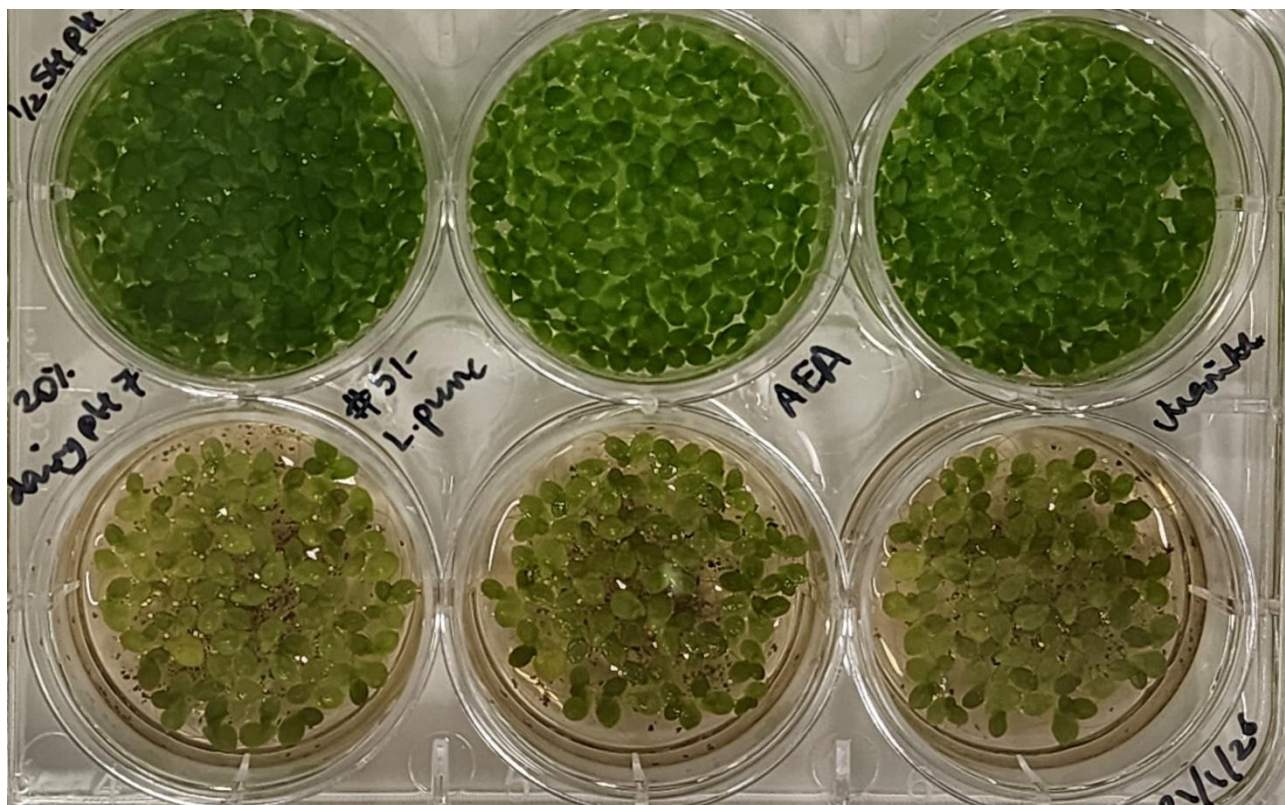
Lemna aequinoctialis



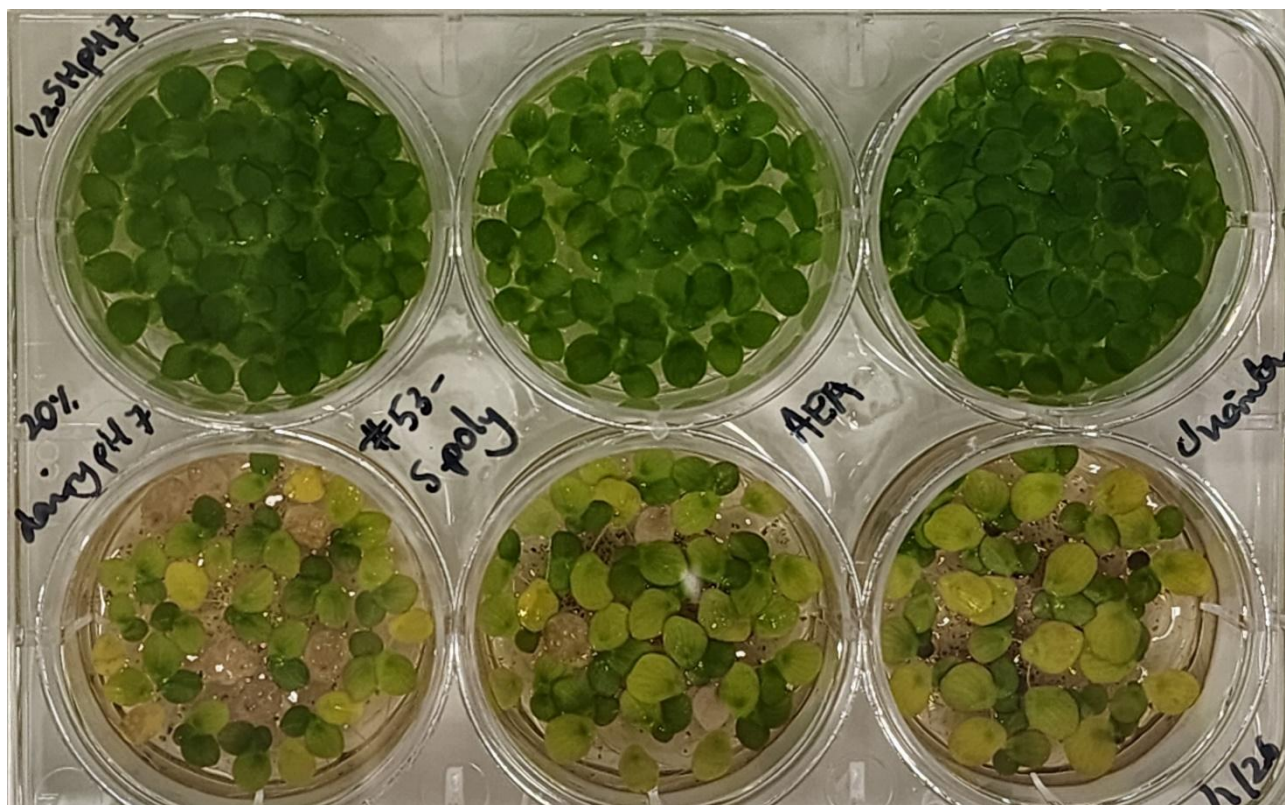
Lemna minuta



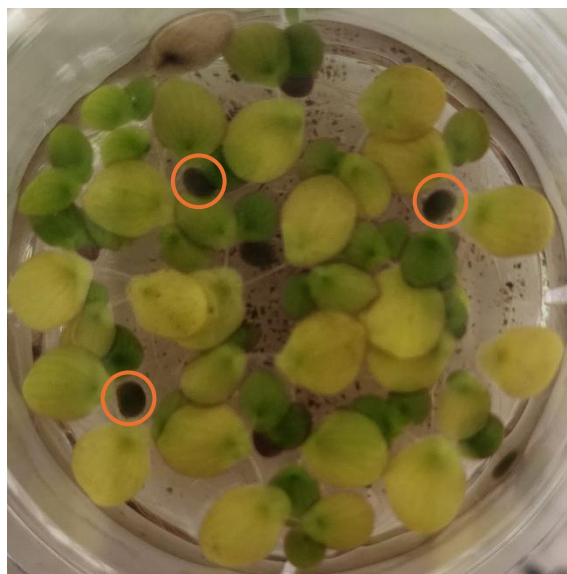
Landoltia punctata



Spirodela polyrhiza



Spirodela polyrhiza – turions shown in orange circles



Duckweed grown in ½ SH show a healthy vibrant green dense mat over the surface. On 20% dairy digestate, growth density and colour intensity is noticeably poorer in all species grown. 20% dairy digestate contains the essential nutrients duckweed requires to grow, however at much lower levels than ½ SH.

Nutrient concentrations (mg/L) in ½ SH medium and 20% dairy digestate compared to optimal growth ranges

Nutrient	Optimal Low	Optimal High	½ SH (control)	20% dairy digestate
Nitrate (N)	3	300	173	ND
Ammonia (N)	20	50	18	41.2
Orthophosphate (P)	0.1	50	40	2.74
Potassium (K)	20	782	483	35.4
Magnesium (Mg)	1.2	240	20	13
Calcium (Ca)	8	800	27	10.8
Sulphate (S)	16	641	26	<10

Optimal Low and High values taken from O'Mahoney et al. 2022. Cultivation of Lemna Minor on Industry-Derived, Anaerobically Digested, Dairy Processing Wastewater

Quick Reference Rule

If the fronds look noticeably paler, thinner, more translucent, or less “solid” compared to your healthy reference stock, treat it as a sign of nutrient deficiency, pH drift, light stress, or emerging contamination.

Additional Signs of Healthy Cultures

- Steady increase in coverage without large bare patches.
- No visible surface films, cloudiness in the media, or foreign particles.
- Minimal dead (white or pale orange) or sinking fronds.

Recommendation: Keep a set of reference photos of healthy cultures for each species under your standard conditions. Weekly photos of your cultures can help track subtle changes over time.

3. System Setup

Recommended Containers

- 90 mm Petri dishes: Ideal for short-term experiments, small-scale assays, imaging, and initial acclimation.
- 250 mL conical (Erlenmeyer) flasks: Preferred for routine stock maintenance and medium-term cultures.
- Larger vessels: 500 mL–1 L flasks, beakers, or trays for scale-up and biomass production prior to non-sterile applications.

Vessel Preparation

Petri dishes (plastic): Supplied pre-sterilised (typically gamma-irradiated) and single-use. Use straight from the sealed pack – do not autoclave, as polystyrene will deform or melt at autoclave temperatures (121°C). Discard after use; do not reuse.

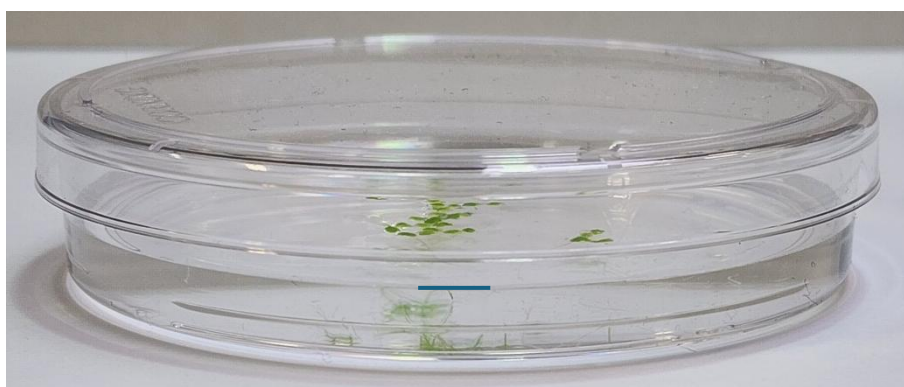
Petri dishes (glass, reusable): Wash and rinse thoroughly with RO/Milli-Q water before autoclaving to remove residue. Autoclave empty flasks at 121°C before first use and before every reuse and allow to cool before adding media.

Flasks (glass, reusable): Wash and rinse thoroughly with RO/Milli-Q water before autoclaving to remove residue. Autoclave empty flasks at 121°C before first use and before every reuse and allow to cool before adding media.

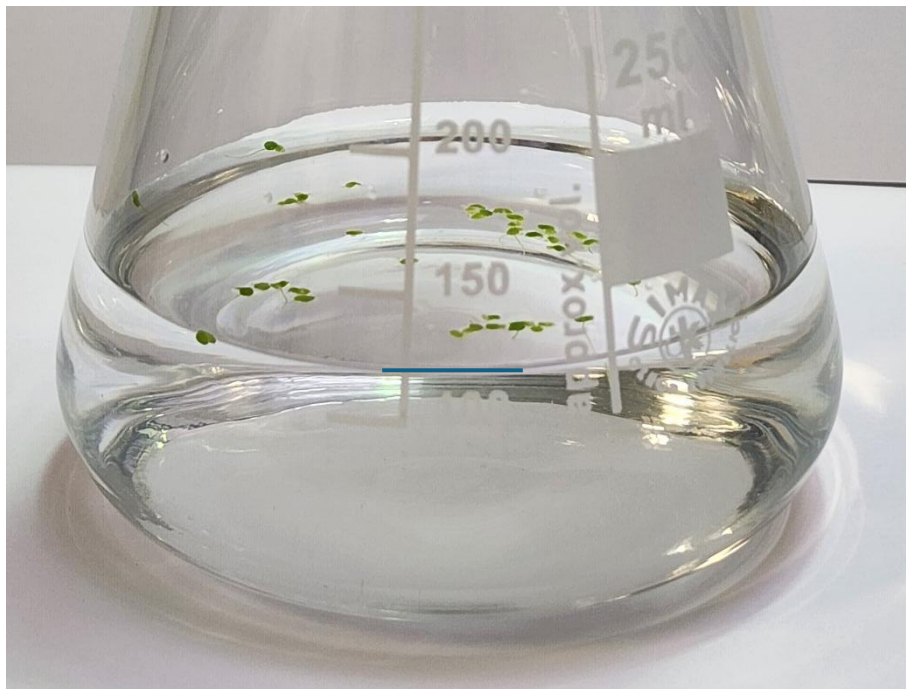
Surface Area vs Volume

Duckweed grows on the water's surface, shallow liquid depth is critical for good light penetration and gas exchange. Aim for at half-filled volumes in Petri dishes and 1-5 cm liquid depth across all vessel types.

- 90 mm Petri dish: 30-50 mL working volume
- 150 mm Petri dish: 80-100 mL working volume
- 250 mL conical flask: 80-120 mL working volume
- 500 mL conical flask: 150-250 mL working volume
- Larger vessels: Scale proportionally while maintaining shallow depth



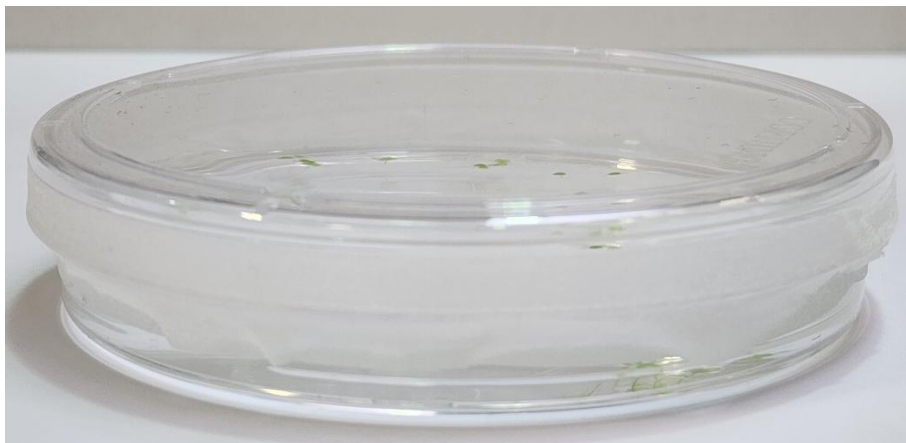
90 mm Petri dish fill level



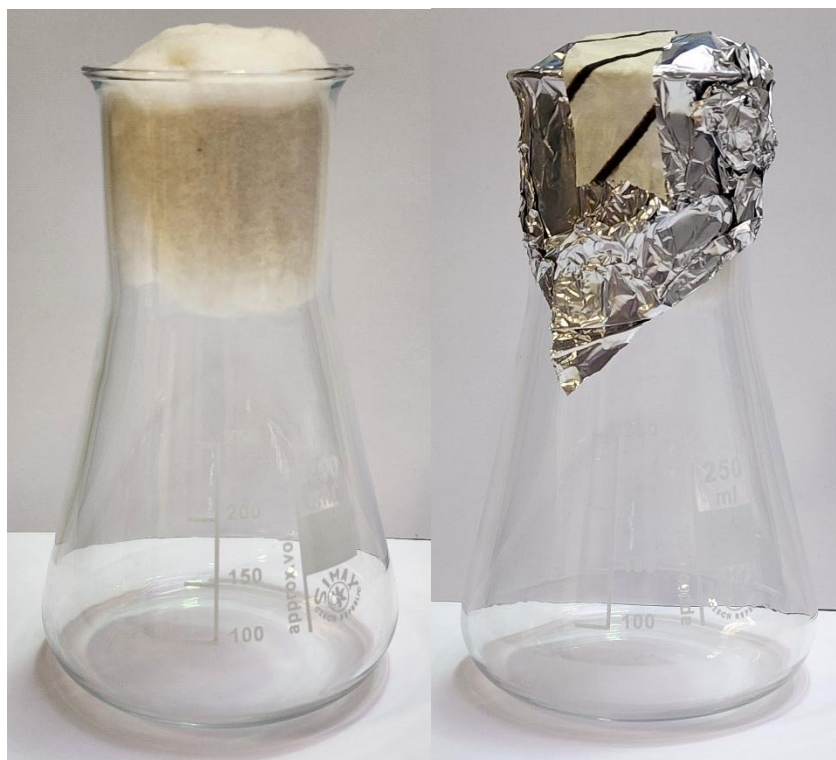
250 mL conical flask fill level

Lid / Closure

- Petri dishes: Secure lids with micropore sealing tape to allow gas exchange while minimising contamination.
- Conical flasks: Use foam or cotton wool plugs inserted in the neck, then cover with aluminium foil. This combination provides good airflow and contamination control.
- Avoid completely airtight seals.



Petri dish sealed with micropore tape



Flask cotton wool plug and aluminium foil cover, post autoclaving

Water Quality

Use reverse osmosis (RO) or Milli-Q (mQ) water for all media preparation and dilutions.

Environmental Conditions – Optimal (Growth Chamber)

- Light: 50–150 $\mu\text{mol m}^{-2} \text{s}^{-1}$ PAR (photosynthetically active radiation), continuous or 16 h light / 8 h dark cycle. Cool-white or full-spectrum LEDs work well.
- Temperature: 20–28°C, most species perform well in this range.
- Humidity: 50–75% (growth chambers like Conviron maintain this automatically).
- Positioning: Even light distribution. Rotate trays periodically if needed.

Bench-top / Non-Chamber Conditions (e.g. for digestate trials)

- Place cultures on a well-lit laboratory bench away from direct sunlight, heat sources, and strong air drafts. (Sunlight can encourage algae growth).
- Use adjustable LED grow lights if natural window light is insufficient.
- Monitor temperature — avoid locations that exceed 30°C or drop below 18°C for extended periods.
- Expect slightly slower and more variable growth compared to a controlled growth chamber.

Important Notes

- Consistent environmental conditions improve reproducibility.
- When transferring cultures from growth chamber conditions to bench-top or non-sterile digestate setups, acclimate gradually over several days where possible.

4. Media Preparation

Standard Medium

½ strength Schenk & Hildebrandt (SH) + 10 mM 2-(N-morpholino)ethanesulfonic acid (MES), pH 6.4

This is the standard medium used for all five species. Half-strength SH provides balanced nutrients, while 10 mM MES buffers the pH effectively ($pK_a \approx 6.1$). The target pH of 6.4 keeps the buffer working well and supports consistent growth across all five duckweed species.

Safety Protection

- Gloves
- Safety glasses
- Lab coat

Materials

- Schenk & Hildebrandt basal salts
- MES hydrate or MES.Na sodium salt
- 1 M HCl (to lower pH)
- 1 M KOH (to raise pH)
- RO or Milli-Q water

Equipment

Analytical balance	Media bottles
Weigh boat	Autoclave tape
Spatula	Autoclave
Beaker	Pasteur pipette
Measuring cylinder	Parafilm
Magnetic stirrer and flea	Scissors
pH meter	

Preparation Steps for 1 L

1. Add ~800 mL RO/Milli-Q water to a 1 L beaker with a stir bar, start the stir bar.
2. Gradually add 1.6 g of Schenk & Hildebrandt basal salts (for half-strength).
3. Add 10 mM MES hydrate or MES.Na (2.13 g of either).
4. Stir on a stir plate until completely dissolved.
5. Place pH probe into the liquid, make sure it's calibrated before you use it.
6. Adjust pH to 6.4 using a Pasteur pipette to transfer either 1 M HCl to lower pH or 1 M KOH to increase pH. *It will depend on which MES is used. MES hydrate has a lower starting pH and will require KOH to increase the pH. MES.Na has a higher starting pH and will require HCl to decrease the pH.*
7. Transfer to a 1 L measuring cylinder, make up to 1 L final volume with RO/Milli-Q water.

8. Cover with parafilm and invert to mix.
9. Transfer 1 L into 2 L Schott bottles or 500 mL into 1 L Schott bottles. *Fill no more than 50% capacity to allow for sufficient headspace during autoclaving.*
10. Label the bottles, open the lid slightly to prevent pressure build up inside the bottle during autoclaving and place autoclave tape on the lid.
11. Autoclave at 121°C for 15–20 min (liquid cycle).
12. Cool to room temperature before use.

Storage

Once cooled and tightly sealed, store media bottles at room temperature on the shelf. Inspect before use and discard if there is any visible contamination, such as mould growth or cloudiness.

5. Core Workflow

This section provides step-by-step procedures for maintaining duckweed cultures. Semi-sterile culture, maintained via a standardised stock revival practice, is the default method for most laboratory work and provides the consistency and reliability needed for reproducible experiments. Non-sterile workflows are used only for applied trials such as digestate or wastewater applications, using semi-sterile stock as the acclimated inoculum source (see Section 1).

If your work requires axenic (sterile) culture, contact the Australian Collection of Duckweed Clones or Jenny Mortimer.

5.1 Starting a Culture

Standardised Stock Revival Practice

To ensure every experiment starts from a consistent, healthy baseline, cultures are revived through a two-stage grow-out from cold storage before use in any experiment.

Materials

- Cold storage stock (source culture)

Equipment

Laminar flow hood	Fine-tipped sterilisable forceps or toothpicks
90 mm Petri dishes or autoclaved 250 mL conical flasks	Fresh autoclaved medium (Section 4)
Micropore sealing tape (dishes) / foam or cotton wool plug + foil (flasks)	Permanent marker (for labelling)
70-80% ethanol spray	

Standardised Stock Revival Practice

1. **Stage 1** – Initial revival (≈ 2 weeks). Remove stock from cold storage and bring to room temperature. Working in a laminar flow hood, aseptically transfer a small number of healthy-looking fronds into fresh, autoclaved medium (Section 4). Grow under standard conditions (Section 3) for approximately two weeks. This stage re-establishes active growth and lets you confirm the revived culture is healthy before relying on it further (see Section 2 for what healthy growth looks like).
2. **Stage 2** – Second grow-out (≈ 2 weeks). Subculture from Stage 1 into fresh medium, again using the laminar flow hood for the transfer. Grow for a further two weeks. This second stage ensures the culture is in a stable, actively growing state – not just recovering from cold storage – before it's used to start experimental work.
3. **Starting the experiment.** Use the Stage 2 culture as the source material for experimental cultures. All experiments within the same series should be started from cultures at the same point in this revival cycle, so that passage history and culture age are consistent across replicates and across time.

This two-stage approach establishes the consistent starting point for all cultures.

When to Return to Cold Storage vs. Continue from an Active Culture

Situation	Action
Starting a new experimental series	Begin a fresh two-stage revival from cold storage
Routine maintenance between experiments within the same series	Continue subculturing from the current active culture (see Section 5.2)
Active culture shows signs of stress, contamination, or drift from expected morphology (Sections 2, 6)	Discard and revive fresh from cold storage – do not attempt to continue
Culture is more than 3 months old	Revive fresh from cold storage, even if the active culture still looks healthy

Aseptic Technique for Transfers

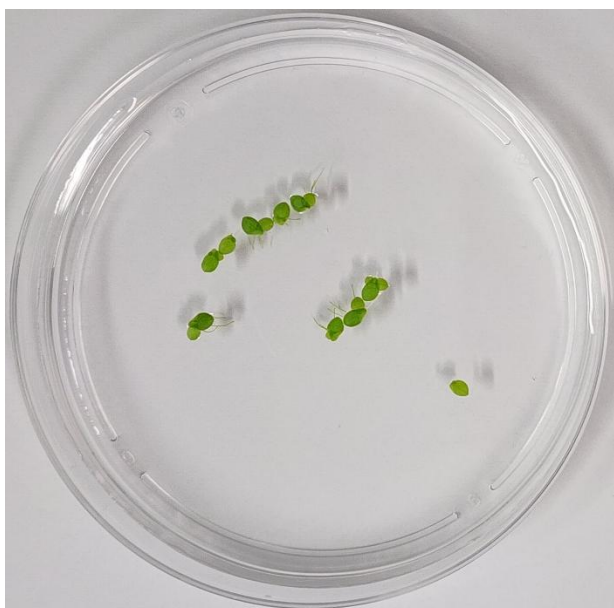
Stage 1 (reviving from cold storage) is always carried out in a laminar flow hood using sterilised tools, regardless of setting. This is the point where contamination risk to your master stock is highest, since it's the direct link back to the Collection's source material.

From Stage 2 onward, laboratory users with hood access should continue using a laminar flow hood and sterilised tools throughout, as this gives the most reliable protection against contamination. Where a laminar flow hood isn't available, clean best-practice handling (a wiped-down surface, sterilised tools, minimising vessel open-time) is an acceptable substitute from Stage 2 onward, though it carries a higher contamination risk than hood-based technique. See Section 5.3 (Non-Sterile/Applied Culture) for how this fits into scaling material toward non-sterile or field conditions.

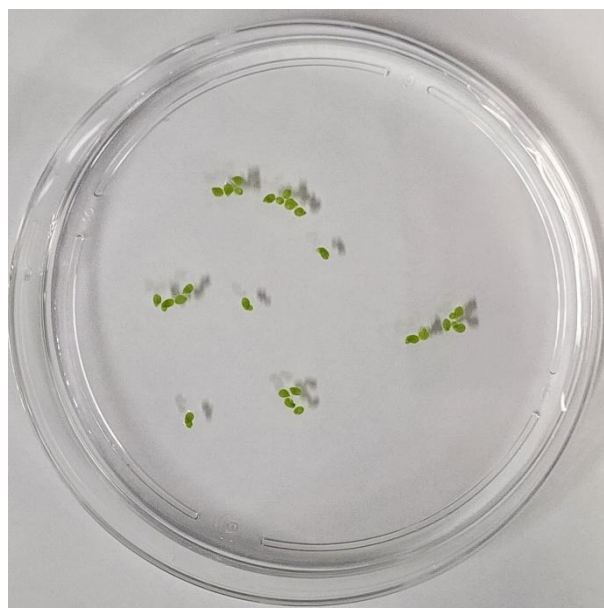
Initial Density

Starting too dense makes it hard to judge whether growth is healthy; starting too sparse risks slow establishment. Aim for a density that supports strong relative growth without crowding, typically 5–25 fronds or 10–50 mg in a 90 mm Petri dish or 250 mL flask at working volume (Section 3). Species with smaller

fronds (e.g. *Wolffia australiana*) will need proportionally more individuals to reach the same starting coverage as larger-fronded species.



Landoltia punctata starting culture, 10 fronds



Lemna minuta starting culture, 25 fronds

5.2 Maintenance and Refresh Cycle

Once a culture is established (Section 5.1), it needs regular observation and periodic subculturing into fresh medium to stay healthy. The process is the same whether you're maintaining background stock or running an experiment – only the timing differs.

Observation

- Weekly visual check against your healthy stock baseline (Section 2) – colour, coverage, and any cloudiness or surface film in the media (Section 6).
- Photograph cultures when starting out. Reference photos are especially useful while you're still building a feel for what healthy growth looks like; less necessary once you're experienced with a species.

Refresh Schedule

Rather than judging media condition case-by-case, cultures are subcultured into fresh medium on a fixed schedule:

Culture Type	Refresh Interval
Stock (background maintenance)	Every 4 weeks
Experimental	Every 2 weeks

The shorter experimental interval keeps cultures in active, consistent growth throughout a series, minimising the nutrient depletion, pH drift, or overcrowding that could otherwise confound results. If coverage approaches full single-layer capacity before the scheduled refresh, subculture early rather than letting the culture become crowded (see Section 2, “Surface Behaviour”).

How to Subculture

1. Working in the laminar flow hood, select healthy fronds from areas of even, vigorous growth – avoid the edges of clumps or crowded patches.
2. Transfer into fresh, autoclaved medium (Section 4) at the standard initial density (Section 5.1).
3. Label the new culture with species, date, and cycle number (subculturing cycles since last cold storage revival), for traceability.
4. Keep the previous culture under observation for a few days to confirm the new subculture has established before discarding it.

Handling Practices

- Use fine-tipped, sterilisable forceps or a sterilised toothpick – avoid touching fronds directly.
- Lift gently from below or the edge; avoid dragging across the surface, which can cause fragmentation or bruising.
- Keep vessels open for the minimum time necessary, and work in the laminar flow hood for anything involving an open vessel.
- If handling multiple cultures in one session, work through them in a consistent order (healthiest/newest first) to reduce cross-contamination risk.

5.3 Non-Sterile (Applied) Culture

Non-sterile culture is used for applied trials where environmental or wastewater microbes are expected and tolerated – primarily digestate or wastewater treatment applications. This mode doesn't replace the standard workflow; it begins from it.

Staged Acclimation

Rather than transferring duckweed directly into a non-sterile target medium, use healthy semi-sterile stock (Sections 5.1–5.2) as the inoculum source and scale it up gradually:

1. Start from an established semi-sterile culture.
2. Scale up biomass using clean technique (Section 5.1, Aseptic Technique for Transfers) until sufficient inoculum is available, hood-based aseptic technique where available, or clean best-practice handling where it isn't.
3. Transfer into the target non-sterile medium, acclimating gradually where possible rather than making an abrupt change (see Section 3, “Important Notes”).

This staged approach improves establishment success compared with introducing duckweed directly into a non-sterile environment.

Harvesting

For applied cropping use, biomass can be harvested from an established culture while leaving the remainder to continue growing. Harvesting doesn't require aseptic handling – standard clean technique is sufficient once a culture is in non-sterile conditions; a laminar flow hood is not necessary at this stage.

Basic principle: Harvest back to approximately 30–40% coverage, leaving enough biomass for the culture to recover and continue growing normally. Harvesting the entire culture isn't recommended.

Gentle handling:

- Use fine-tipped forceps to lift fronds out, rather than pouring, scooping, or agitating the culture.

- Harvest from a consistent area rather than scattered picking, to minimise disturbance to the remaining culture.
- Smaller or more delicate species (e.g. *Wolffia australiana*) fragment more easily than larger-fronded species – handle with extra care.

6. Troubleshooting

Symptom	Likely Cause(s)	Action
Slow or no growth	Nutrient depletion (media overdue for refresh); low light; temperature outside optimal range; overcrowding	Check refresh schedule (Section 5.2); verify light/temp against Section 3; subculture if overdue or overcrowded
Pale, thin, or translucent fronds	Nutrient deficiency; light stress; early contamination; pH drift	Compare against Section 2 reference cues; refresh media early; inspect for early contamination signs below
Overgrowth then crash	Culture left too long between refreshes; overcrowding blocking light/gas exchange	Refresh on schedule (every 4 weeks per Section 5.2) rather than waiting for visible crowding
Cloudy media or surface film	Contamination (bacterial or fungal)	Isolate culture; do not use for experiments; revive fresh from stock (Section 5.1); if persistent, review handling technique
Unexpected particles or discolouration in media	Contamination; degraded media components	Discard media; prepare fresh batch (Section 4); check media storage conditions and shelf life
Clumping or uneven distribution	Natural species tendency (Spirodela, Landoltia); insufficient surface area; disturbance during handling	Compare against expected species behaviour (Section 2); check vessel surface area (Section 3); minimise unnecessary handling
Sudden stagnation despite normal conditions	Subtle contamination not yet visible; pH drift; equipment issue (light, temperature)	Weekly photo comparison (Section 5.2) to catch early drift; verify equipment; consider early refresh

7. Disposal of Cultures and Waste Media

Several of the species covered in this manual are fast-spreading and not native to all environments. Cultures and waste media must never be released into drains, waterways, or the general environment – always kill material before disposal.

Standard Method: Bleach Treatment

Bleach is the recommended disposal method for all cultures, as it reliably kills both actively growing fronds and dormant resting structures (turions), which some species can produce, notably *Spirodela polyrhiza*. Turions are specifically adapted to survive cold conditions, so cold-based methods alone cannot be relied on for certain kill.

1. Drain excess liquid from the culture vessel.
2. Add a bleach solution of 1% and allow to stand overnight, ensuring all fronds are fully submerged.
3. Dispose of the treated material and solution according to your lab's standard chemical waste procedure.

Bleach treatment is mandatory (not optional) for any PC2-classified material, regardless of species or apparent culture health.

Alternative: Freezing

Placing drained cultures in a standard freezer (– 20°C) for two weeks is likely to kill actively growing fronds, but is not a verified method for killing turions, which are adapted to survive cold conditions in nature. If freezing is used in place of bleach, confirm effectiveness by thawing a sample and holding it under normal growth conditions for one to two weeks to check nothing sprouts, before relying on this method routinely.

Quick Reference Guide

A condensed lookup for readers already familiar with the full manual. See the referenced sections for full detail.

Vessel Prep (Section 3)

- Plastic Petri dishes: pre-sterile, single-use, never autoclave
- Glass Petri dishes: autoclave before every use
- Glass flasks: autoclave before every use, include foam or cotton wool plug and foil to cover the opening

Working Volumes (Section 3)

- 90 mm Petri dish: 30-50 mL
- 250 mL conical flask: 80-120 mL

Environmental Conditions (Section 3)

- Light: 50–150 $\mu\text{mol m}^{-2} \text{s}^{-1}$ PAR, continuous or 16 h/8 h
- Temperature: 20–28°C

Standard Medium (Section 4)

- ½ strength Schenk & Hildebrandt + 10 mM MES, pH 6.4
- Autoclave at 121°C for 15–20 min

Revival Cycle (Section 5.1)

- Stage 1 (from cold storage): ~2 weeks
- Stage 2 (second grow-out): ~2 weeks
- Initial density: 5–25 fronds, 10 – 50 mg]

Refresh Schedule (Section 5.2)

- Stock cultures: every 4 weeks
- Experimental cultures: every 2 weeks

Harvesting – Non-Sterile/Applied (Section 5.3)

- Harvest back to ~30–40% coverage

Disposal (Section 7)

- Bleach treatment is standard for all material; mandatory for PC2 material
- Never release cultures into drains or waterways

Troubleshooting Quick Index (Section 6)

- Slow/no growth → check refresh schedule, light, temperature
- Pale/translucent fronds → nutrient/light stress or early contamination
- Cloudy media/film → contamination; isolate and revive fresh from stock
- Clumping/uneven distribution → species tendency or handling disturbance